

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 30-316-1
Fire Suppression Pump System	16-09-2025

EDMS 30-316-1

SPECIFICATION

FOR

STATIONARY PUMPS FOR FIRE PROTECTION SYSTEM

G.p.m – bar

Issue: Feb.-2026/ Rev- 1

- This revision contains option items that must be selected by EDC before bidding.

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1- SCOPE

This specification covers the minimum technical requirements for design, manufacturing, supply, installation, inspection, testing and delivery of fire suppression pump system for supplying the liquid of fire protection. The purpose is to provide reasonable degree of protection for life and property from fire through installation requirements for stationary pumps for fire protection.

2- APPLICABLE STANDARDS

The equipment/material covered in this specification shall comply with the latest edition/amendment of the standards given in Table (1). Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply. Any deviations from the listed standards or the provisions of this specification shall be clearly set out in the offer.

Table (1)

Standard No.	Description
NFPA20	This standard shall apply to centrifugal single-stage and multistage pumps. Requirements are established for the design single-stage and multistage pumps, pumps drivers, and associated equipment.
Egyptian fire code	Section 4
NFPA 70E	Standard for Electrical Safety in Workplace
NFPA20 HANDBOOK	Stationary Fire Pumps

3- CONSTRUCTION REQUIREMENTS

- The required fire pump package shall be designed and manufactured according to NFPA20
- The package shall consist of:**
 - 1- Electrical fire pump set including but not limited to the fire water pump,

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approved electrical motor, control unit, fabrication.

- 2- Diesel fire pump set including but not limited to the fire water pump, approved diesel engine, control unit, fabrication.
 - 3- Electrical motor driven fire water jokey pump set with control panel.
- The pump shall be centrifugal, horizontal split case pump, or end suction top discharge pump, or in line pump, or turbine pump (The selection shall be according to the required system based on the hydraulic calculations of a fire consultant approved by the Civil Protection Department).

3.1 Pump Performance:

- The pump performance shall be according to NFPA20 (UL-Listed FM-Approved is option according to ----- EDC requirement).
- Provide pressure and flow rate according to the design point.
- Shall furnish not less than 150 percent of rated capacity at not less than 65 percent of total rated head.
- Shutoff head shall not exceed 140 percent of rated head of the pump.
- Service factor at which a motor shall be used not exceed 1.15-The maximum motor load may occur at pump outputs at or above 150% percent flow rate (as explained in NFPA 20, NFPA 20 HANDBOOK).
- Fire pumps shall be dedicated to and listed for fire protection service.
- The available flow and pressure at fire pump discharge shall be adequate to meet the maximum fire protection demand.
- Stored supply plus reliable automatic refill shall be sufficient to meet the demand placed upon it for the design duration.

3.2 Electric drive for pumps:

- Acceptable drivers for pumps at single installation shall be electric motors.

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- Pump shall not be equipped with more than one driver.
- The motor should be UL-listed / FM-approved (Option according to ----- EDC requirement).
- **Pump material** shall be as follows:
 - Casing: ductile cast iron
 - Impeller: bronze
 - Shaft: stainless steel 420
 - Casing/impeller replaceable wearing ring: bronze
 - Coupling shall be flexible type with non-sparking guard material.
- Bidder shall provide approved **control panel** according to the following:
- UL Listed FM Approved (Option according to ----- EDC requirement).
 - Suitable for: 380v/3ph/50Hz
 - Stander features according to NFPA20.
 - Touch Screen
 - Nema 2 if enclosure **ODP**. / Nema 3 if enclosure **TEFC**.
 - Star delta start open transition.
 - (Skid mounted) (Pump, driver, controller fixed in one same skid).
- **Electric motor** shall be according to the following:
- The motor should be UL listed (Option according to ----- EDC requirement).
 - Suitable for: 380 V / 3ph / 50 Hz.
 - Service factor: 1.15
 - Enclosure: **ODP** (PEN DRIP PROOF) (**TEFC** is option according to EDC requirements).
 - Temperature rise class B CL: B
 - Insulation class: CL: F

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- Lubrication: greaser
- **Pump Accessories:**
 - Automatic air- release valve
 - Casing circulation automatic relief valve
 - Suction pressure gauge
 - Discharge pressure gauge filled with glycerin
- **Valves:**
 - 2 OS&Y valves at suction and discharge
 - 1 check valve at discharge only

3. 3 **Diesel engine drive for pumps:**

- The pump shall be centrifugal
- UL listed / FM approved
- **Pump material shall be as follow:-**
 - -casing ductile iron
 - -impeller bronze
 - -shaft stainless still 420
 - -casing/impeller replaceable wearing ring: bronze
 - -coupling shall be flexible type with non-sparking guard material.
- **Pump Accessories:**
 - Automatic air- release valve
 - Casing circulation automatic relief valve
 - Suction pressure gauge
 - Discharge pressure gauge filled with glycerin
 - Separate Diesel fuel tank with capacity according to Nfpa20

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- Pump, driver, controller shall be mounted on the same skid.
- **Diesel engine according to the following:**
 - **Diesel Motor**
 - UL Listed –FM Approved (Option according to ----- EDC requirement).
 - Diesel engine- four strokes, water cooling or air cooling
 - Cooling system closed circuit by heat exchanger
 - Working manually and automatically
 - Diesel engines for fire pump drive shall be of the compression ignition type
 - The power not less than 10% greater than max. Horsepower required by the pump at the end of the curve
 - **Engine accessories**
 - (1) Flexible exhausted
 - (1) Exhausted silencer: residential type with spark arrestor
 - (1) Dual batteries: lead acid storage type providing 100% stand by reserve capacity
 - Fuel system according to Nfpa20
 - (1) Fuel store tank not less than 1.1 gal/hp
 - Fresh water engine cooling system: factory installed water, water piping valve and coolant pump by pass piping.
 - **Engine controller**
 - The automatic engine controller must be approved specifically for pump service the controller must be containing the following feature:
 - UL Listed – FM Approved (Option according to ----- EDC requirement).
 - Built in battery charger
 - Time – clock for weekly automatic

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- Power failure start
 - Low fuel level switch
 - Engine shall be provided with an over speed shut - down device
 - Touch Screen.
 - Nema 2 if enclosure **ODP** / Nema 3 if enclosure **TEFC**.
- **Instrument panel**
- All engine instruments shall be placed on panel secured to the engine or inside an engine base plate – mounted controller
 - Engine rev. counter: - tachometer or other means shall be provided to indicate revolutions per minute of engine including zero, at all time
 - Oil pressure
 - Engine shall be provided with an oil pressure gauge or other means to indicate lubricating oil pressure
 - Temperature
 - Engines shall be provided with temperature gauge or other means to indicate engine coolant temperature at all times
 - The Engine shall be provide with means to indicate the type of trouble being signaled in (fuel injection trouble – low fuel pressure – any primary sensor failure).
- **Valves**
- 2 OS&Y valves at suction and discharge
 - 1 check valve at discharge only
 - 1 Pressure relief at discharge only
 - Tamper switch
 - Flow meter

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3.4 Jokey Pump

- The pump shall be centrifuge, vertical, multi stage
- The pump shall be rated for to gpm@bar differential head.
- The pump shall be completed with stander automatic pressure controller including high/low pressure switch
- The head must be 5% to 10% at design head.

- **Electric motor shall be according to the following:**

- Suitable for: 380 V / 3ph / 50 Hz
- Service factor 1.0
- Enclosure: **ODP** (PEN DRIP PROOF) (**TEFC** is option according to EDC requirements).
- Temperature rise CL: B
- Insulation class: CL: F
- Lubrication: greaser
- Pump control panel D.O.L. starter
- Seal: mechanical
- Body of pump: stainless steal
- Impeller stainless steal
- Shaft stainless steal
- The type of control panel is the same the type of main panel

3.5 Discharge header

- UL/FM Flow meter shall be insulation at discharge header (Option according to ---
-- EDC requirement).

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4- GENERAL REQUIREMENTS

- Certificate of origin and the Euro1 certificate must be submitted.
- Pumps must have an agent in Egypt, and a (س٤١) certificate must be attached.
- Availability of a maintenance center, after-sales service, and original spare parts.
- Providing factory test certificates for all pumps and UL-FM certificates.
- Submitting a customs clearance certificate for pumps.
- Pumps must be newly manufactured, not more than 3 years old.
- Warranty against manufacturing defects for at least one year from the delivery date. The warranty is extended by an additional year if the supplier has not previously supplied.
- The supplier installs and tests the equipment according to the operating curves. All works should be delivered to the Civil Protection Department.
- Original catalogs and curves of the pumps should be submitted with the offer. Similar Original catalogs and curves should be delivered with the system.
- Valves must be UL Listed - FM Approved, with a prove certificate (Option according to ----- EDC requirement).
- Control panels for each pump must be UL Listed - FM Approved, with a prove certificate. (Option according to ----- EDC requirement).
- The electric motor must be UL Listed, with a prove certificate, (Option according to ----- EDC requirement).
- The diesel motor must be UL Listed FM Approved, with a prove certificate. (Option according to ----- EDC requirement).